

In addition, these processes drew a substantial amount of finances.

To address these challenges, the department collaborated with Lipi Data Systems Ltd to deploy self-service kiosks across 10,000 *gram panchayats* in Rajasthan. The kiosks, called e-Mitra Plus, were rolled out in a phased manner. The first phase started in October 2017, and launched 400 services across all the 10,000 kiosks in the rural areas of the state. The idea was to reduce the hassles that citizens faced while collecting documents, by making these available in close proximity.

The e-Mitra Plus setup

Each self-service kiosk consists of two screens. The one at the side is used for various kinds of communication like video-conferencing, live video broadcasts and advertisements. Ajay Jha, general manager - sales, Lipi Data Systems, says, "The screen is often used to broadcast public announcements or important service-related messages by the government. Customised ads can also be broadcast."

"The screen has a two-way communication facility, using which authorised administrators from different locations can log in to the system and communicate using the screen in case of an emergency." Integrated speakers and mics enable two-way communication.

The other face to the front of the system is the main transaction unit.



e-Mitra Plus kiosk

Here, selected official documents can be generated. Inputs are given through simple button presses or touchscreen control. An integrated printer prints the documents on the spot.

The applicant has to submit a request for documents at a nearby office. On verification, either an existing identification number (such as Aadhaar number, PAN, etc) or OTP

is provided to the applicant. This is used as input for the kiosk to print the corresponding document. Charges may be associated with the service as set by the government. Payment can be done using the kiosk through different modes such as cash, debit card, credit card or any other method.

Kiosks installed in the first phase are capable of printing certificate documents only, like birth, land and education.

In the next phase, new kiosks will be installed that will come with card printing facility, too.

Four types of cards can be printed in a single kiosk, as per specified requirements (like Aadhaar card, PAN card, driving licence and so on). Moreover, the kiosks can also display transaction or application statuses. While Lipi Data Systems delivered the kiosks, network provider Rajnet set up the optical-fibre infrastructure for connectivity. Software service providers were involved in developing the backend operating platform.

Benefits achieved

Without human intervention, errors and delays have reduced substantially. Jha adds, "This process helps avoid the various channels associated in manual procurement of documents. Applicants can avoid long queues and multiple visits to the offices. Conversely, service offices can work quickly with reduced work load. All this leads to improved citizen satisfaction and transparent e-governance."

The digital display helps create mass awareness starting from rural regions. Importance of documents, necessary procedures and regulations are made familiar to the masses. Announcement broadcasts are better perceived among the public. Instant video-conferencing feature enables quick communication across distributed teams.

Also, a lot of money is saved. Jha says, "Automated kiosks reduce the requirement of investing in additional manpower, separate counters and office setups just for handing documents."

Telecom providers like Jio, Airtel and Vodafone are also collaborating for this project. Kiosk-based bill payments are enabling all associated organisations to generate revenues as well.

Cost points

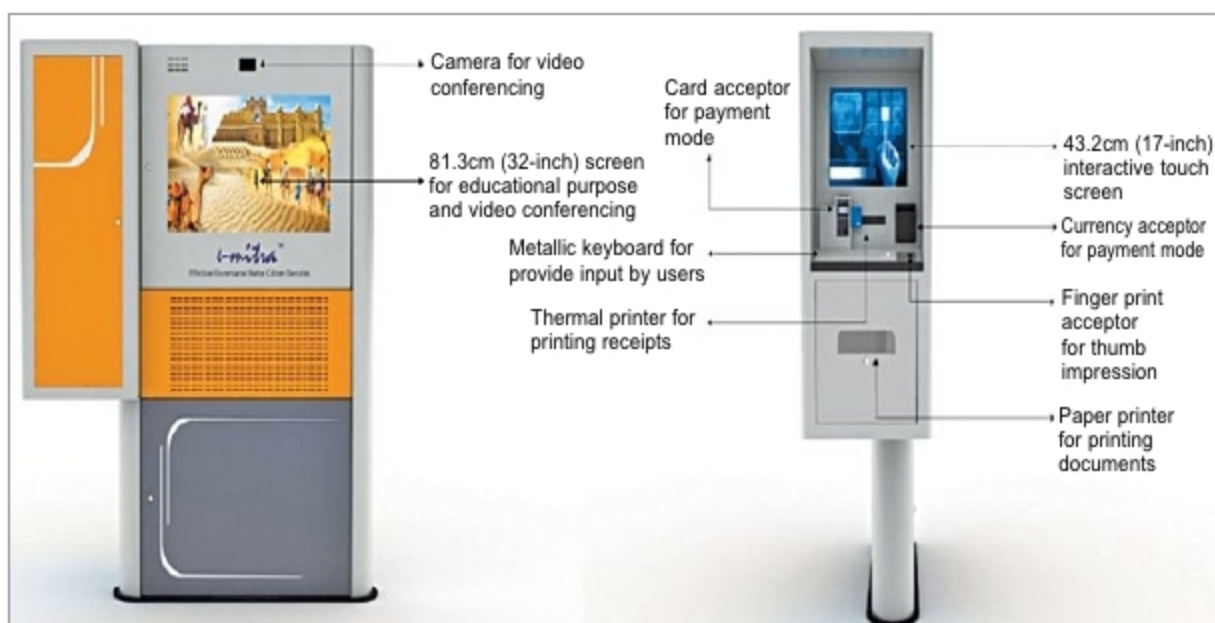
Jha mentions that price of each kiosk is about ₹ 200,000. Investments were required in areas like extensive software development, network setup and other infrastructural requirements. Logistics to remote areas and setup processes also came with challenges and financial requirements. The complete cost of the project at the state scale is over ₹ 3 billion.

Going forward

The next phase of kiosk installation will be visible within the next three months. Five thousand new kiosks will be installed throughout the state of Rajasthan, including major urban areas.

The feedback has been positive from citizens and government officers alike. Local language support and well-instructed screens enable easy usage for commoners. This kind of a solution can be a great example for the rest of India to ease the process of document procurement for Indians and give a big boost to Digital India initiative. **EFY**

—Paromik Chakraborty, technical journalist, EFY



Features of e-Mitra Plus kiosk